

Results

The proposed Smart Waste Management System was successfully simulated using Wokwi.

The ultrasonic sensor correctly detected different garbage levels by changing the simulated distance.

The Raspberry Pi Pico W calculated the fill percentage accurately.

Sensor data was transmitted successfully to the ThingSpeak IoT dashboard.

The dashboard displayed real-time values and graphical trends.

When the garbage level exceeded 80%, the system displayed the dustbin full status.

The simulation verified the correctness of the proposed IIoT solution without using physical hardware.

Conclusion

The Smart Waste Management System demonstrates how Industrial Internet of Things (IIoT) technology can improve waste monitoring and collection.

Using Raspberry Pi Pico W, an ultrasonic sensor, and cloud-based monitoring, the system provides real-time information regarding the fill level of a dustbin.

The project was successfully designed and verified using the Wokwi simulation platform, proving that the concept can be implemented in practical smart city applications.